

ZACH PARK

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EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering - Research

Pittsburgh, PA

May 2028

- Current Coursework: Robot Dynamics and Analysis, Machine Learning and AI, Trustworthy AI, Modern Control Theory

The Cooper Union for the Advancement of Science and Art

GPA: 3.86/4.0 (Summa Cum Laude) | New York, NY

Bachelor of Engineering in Mechanical Engineering, Minor in Bioengineering

May 2026

- Related Coursework: Mechatronics, Signal Processing, Design & Prototyping, Materials Science, Mechanical Vibrations

PROFESSIONAL EXPERIENCE

Product Design Intern, HUROTICS Inc.

Seoul, South Korea | Jun. 2026 – Aug. 2026

- Designed company's first gait test robot in CAD with LabVIEW-based actuator control architecture, incorporating torque, backlash, and encoder feedback for automated exosuit validation, resulting in company-funded \$10K production build
- Collaborated with apparel and embedded teams to design core mechanical systems for company's next-generation backpack exosuit in SolidWorks, with multiple components selected for final production design
- Engineered ToF-IMU gait-sensing system from sensor hardware and Teensy firmware through signal processing, feature extraction, and ML-based classification in MATLAB, improving company's existing gait-labeling accuracy by 13%
- Developed LiDAR-camera-IMU perception pipeline in Python using multimodal sensor fusion, optical-flow tracking, depth-profile analysis, and regression-based prediction, detecting upcoming terrain 34% earlier than previous sensing system

Mechanical Design Intern, Mount Sinai Hospital BioDesign

New York, NY | Aug. 2025 – May 2026

- Prototyped reusable silicone intracerebral hemorrhage (ICH) brain-and-skull phantom in Fusion 360 for neurosurgical training and device validation, replicating tissue feel while reducing fabrication time by 75% and material cost by 40%
- Created patient-specific chronic subdural hematoma (CSDH) testbed from computed tomography anatomy with <1 mm cranial deviation, replaceable dura, silicone brain, and configurable burr-hole access for device testing
- Refined cranial access geometry across 4 design iterations in collaboration with neurosurgeons, engineers, and medical-device industry partners, increasing endoscopic field of view by 17%

PROJECT EXPERIENCE

Cable-Driven Tail for Quadruped Balance Recovery, Cooper Union

New York, NY | Jan. 2026 – May 2026

- Simulated and designed cable-driven modular robotic tail for quadruped balance recovery, modeling simplified robot-tail dynamics in Python and deriving kinematics for 1-, 2-, and 3-segment configurations before hardware testing
- Implemented IMU-feedback PD control in C++ on an Arduino-based microcontroller to command antagonistic tail actuators and used ROS 2 to control quadruped joint motion during physical experiments
- Analyzed IMU response data in Python across 18 controlled disturbance trials, comparing active and passive tail configurations and achieving damping ratios up to 1.66x passive baseline

Active Armrest Robot for Microsurgery Clinicians, Cooper Union

New York, NY | Aug. 2025 – May 2026

- Led 3-engineer interdisciplinary team with Mount Sinai clinicians to develop stool-mounted active armrest for ENT surgeons, integrating C++ control, machining, lead-screw actuation, and ToF sensing for hands-free height adjustment
- Validated armrest through sEMG signal processing, DWPose posture tracking, and drawing-error analysis, reducing trapezius EMG RMS by 30%, drawing deviation by 23%, and forward-head offset by 11%

Wearable Sensor for Dialysis Access Monitoring, Cooper Union

New York, NY | Jan. 2025 – May 2026

- Partnered with Mount Sinai nephrologists to engineer dual-axis piezoelectric wearable for quantitative arteriovenous fistula (AVF) vibration monitoring, combining orthogonal sensing and LabVIEW-based signal processing
- Fabricated <\$200 vascular test rig with custom pulsator and compliance chamber, replacing \$3,000+ pump, supporting 30+ validation trials, and contributing to **2nd Place finish and \$15K award in 2025 NIH DEBUT Challenge**

SKILLS

CAD & Engineering: SolidWorks, Fusion 360, ANSYS, LS-DYNA, MATLAB, Simulink, NI LabVIEW

Programming & Robotics: Python, C/C++, ROS 2, PyTorch, OpenCV, Git